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Does Finance Benefit Society? A Language Embedding Approach

Contents

1 Big picture

- **Question.** How has popular sentiment toward finance varied across countries and over time, and does that sentiment relate to financial development, crises, and economic growth?
- **Core idea.** The paper treats books as a long-run record of how society discusses finance. Since surveys do not exist for people living in the late nineteenth and early twentieth centuries, the authors use millions of books to infer historical public sentiment toward finance.
- **Method.** The authors apply BERT, a large language model, to finance-mentioning five-word sequences from the Google Books Ngram Corpus. They embed each finance sentence, project it onto a positive-minus-negative finance dimension, and aggregate those scores by language-year.
- **Sample.** The main panel covers eight country-language pairs from 1870 to 2009: American English/United States, British English/United Kingdom, Chinese/China, French/France, German/Germany, Italian/Italy, Russian/Russia, and Spanish/Spain. The Chinese sample starts later because early Chinese finance mentions are sparse.
- **Main descriptive fact.** Finance sentiment differs persistently across languages. U.S. and U.K. English are the most positive; Spanish, French, Chinese, and Italian are in the middle; German and Russian are negative. The ordering is remarkably stable despite meaningful within-country time-series variation.
- **Capitalism result.** Languages associated with more capitalist countries tend to discuss finance more positively. Finance sentiment is higher where governments are more right-leaning, citizens express stronger preference for private ownership of business, and starting a business is easier.
- **External validation.** Finance sentiment is positively related to direct measures of financial participation: equity holdings by households, household loans/GDP, and nonfinancial private-sector loans/GDP. It is negatively related to income inequality.
- **Placebo logic.** The authors repeat the measurement exercise for other industries and a neutral word. Sentiment toward coal, paper, tobacco, and January does not display the same country ordering or trend, suggesting that the main measure is finance-specific rather than general textual positivity.
- **Financial-crisis fact.** Finance sentiment falls before, not after, banking distress episodes. The decline occurs one year before bank equity declines and remains predictive even after controlling for credit growth.
- **Macroeconomic fact.** Positive shocks to finance sentiment are followed by higher output growth and higher credit growth. A 1 percentage point finance-sentiment shock is associated with a roughly 0.3 percentage point increase in GDP growth several years later and about a 0.4 percentage point increase in credit growth in the credit-data subsample.
- **Nuance.** Short-run sentiment improvements are followed by growth, but prolonged sentiment expansions are negatively correlated with future GDP growth after controlling for credit expansions. This helps reconcile the paper with the credit-boom literature.
- **Main contribution.** The paper provides a long-run, cross-country, text-based measure of finance sentiment and connects it to financial participation, inequality, crises, credit, and output.

- **Takeaway.** Popular beliefs about finance appear to be persistent, country-specific, and economically meaningful. The evidence is not a clean causal experiment, but it strongly suggests that attitudes toward finance are an important part of financial and macroeconomic history.

2 Data and measurement (key objects)

2.1 Why this setting is useful

- The key obstacle is that trust in finance and sentiment toward finance are usually measured with modern surveys, while the paper wants a historical panel covering more than a century.
- Books provide a long and comparable textual archive. The Google Books Ngram Corpus contains yearly usage frequencies for words and phrases from millions of books.
- The corpus allows the authors to ask how finance was discussed in different languages, in different years, and around major historical events such as revolutions, wars, financial panics, and regime changes.
- The setting is useful because finance sentiment may affect both sides of financial markets: households' and firms' demand for financial services, and political support for financial regulation or financial development.

Interpretation. The paper is not measuring the private opinions of every household. It is measuring the tone of published language around finance. This may represent the broader public, or at least a historically influential literary and political elite.

2.2 Textual data

The text data come from the 2012 Google Books Ngram Corpus.

- **Unit of observation in text:** five-word sequences, or 5-grams, that contain the stem of the word "finance" in a given language.
- **Languages:** American English, British English, Simplified Chinese, French, German, Italian, Russian, and Spanish.
- **Period:** 1870–2009 for most languages. Chinese is shorter because pre-1922 finance mentions are sparse.
- **Why 5-grams:** sentiment depends on context. A five-word sentence gives BERT enough local context to distinguish, for example, financial support from financial panic.
- **Preprocessing:** the authors strip case, symbols, double spaces, part-of-speech tags, and positional tags, then extract finance-mentioning sentences.

Interpretation. The paper uses short book excerpts as repeated historical observations of how authors put finance into context.

2.3 Finance mentions across languages

The number of finance-mentioning sentences differs substantially by language.

- American English has about 220 thousand unique finance sentences and 79 million total finance sentence occurrences.
- British English has about 48 thousand unique sentences and 15 million total occurrences.
- Simplified Chinese has about 196 thousand unique sentences and 305 million total occurrences.
- French has about 100 thousand unique sentences and 43 million total occurrences.
- German has about 28 thousand unique sentences and 7 million total occurrences.
- Italian has about 23 thousand unique sentences and 9 million total occurrences.
- Russian has about 187 thousand unique sentences and 250 million total occurrences.
- Spanish has about 89 thousand unique sentences and 33 million total occurrences.

Interpretation. Some languages have many more finance mentions than others, so the measurement precision differs across countries and over time. This is one reason the authors focus on 1870 onward and treat early Chinese data carefully.

2.4 Macroeconomic and financial data

The paper links finance sentiment to macroeconomic and financial outcomes.

- **GDP and credit data:** the authors use the macrohistory data set of Jorda, Schularick, and Taylor for advanced economies. For China and Russia, they supplement GDP and population data from Barro-Ursua.
- **Credit measure:** total loans to the nonfinancial private sector, used as a proxy for aggregate credit availability.
- **Credit sample:** credit-growth data are available for six countries: France, Germany, Italy, Spain, the United Kingdom, and the United States. China and Russia are excluded in credit-based regressions.
- **GDP growth:** annual growth in real per capita GDP.
- **Banking distress:** banking crises and bank-equity decline measures come from Baron, Verner, and Xiong, with robustness checks using alternative crisis chronologies.
- **Financial participation:** equity holdings by households come from OECD data; household loans and private-sector loans are from macrohistory data.
- **Income inequality:** the Gini coefficient comes from the World Income Inequality Database.

Interpretation. The paper combines a new text-based sentiment panel with standard historical macro-financial data to test whether the text measure behaves like a meaningful economic variable.

2.5 BERT embeddings

The authors use BERT because it produces contextualized sentence embeddings.

- BERT is pre-trained on large text corpora, including BooksCorpus and Wikipedia. This transfer-learning feature is valuable because some historical language-year cells contain relatively few finance mentions.
- BERT processes words in relation to all other words in the sentence and is bidirectional, so it can distinguish different contexts for the same word.
- The authors use BERT Base for American and British English, BERT Base Chinese for Simplified Chinese, and BERT Base Multilingual Cased for French, German, Italian, Russian, and Spanish.
- BERT adds a special [CLS] token at the start of each sentence. The embedding for this token is used as the sentence embedding.

Interpretation. BERT allows the measure to go beyond dictionary counting. The sentence "financial support of the center" and the sentence "financial panic swept the country" both contain finance-related words, but BERT places them in very different regions of embedding space.

2.6 Positive-minus-negative finance dimension

The paper defines a language-specific finance positivity dimension using pairs of positive and negative benchmark sentences. For English, examples include:

Positive finance sentence	Negative finance sentence
financial services benefit society	financial services damage society
finance is good for society	finance is bad for society
finance professionals are mostly good people	finance professionals are mostly corrupt people
finance positively impacts our world	finance negatively impacts our world
financial system helps the economy	financial system hurts the economy

For each language, the authors translate these sentence pairs. Let p_i denote the positive-minus-negative embedding for language i. It is constructed by averaging the vector differences between the positive and negative benchmark sentences.

Interpretation. The benchmark sentences define the direction in embedding space that corresponds to "finance benefits society" rather than "finance harms society."

2.7 Sentence-level finance sentiment

For each finance-mentioning sentence j in language i, with sentence embedding s_{ji}, the paper defines sentiment as cosine similarity with respect to the positivity vector p_i:

$$a_{ji} = \frac{s_{ji} \cdot p_i}{\|s_{ji}\|\|p_i\|} = \frac{\sum_d s_{jid} p_{id}}{\sqrt{\sum_d s_{jid}^2}\sqrt{\sum_d p_{id}^2}}. \tag{1}$$

- $a_{ji} > 0$: the sentence places finance closer to the positive direction.
- $a_{ji} < 0$: the sentence places finance closer to the negative direction.
- $a_{ji} \approx 0$: the sentence is roughly neutral along the finance-sentiment dimension.

Interpretation. The method measures the angle between a finance sentence and the positive finance direction. A more positive sentence has a smaller angle with the positive dimension and therefore a higher cosine similarity.

2.8 Annual finance sentiment

The annual sentiment index for language i in year t is the frequency-weighted average of sentence-level sentiment scores:

$$f_{it} = \sum_j a_{ji} \times \frac{c_{jit}}{\sum_k c_{kit}}, \tag{2}$$

where c_{jit} is the number of times sentence j occurs in language i and year t.

- The sentence sentiment score a_{ji} is fixed over time.
- Time-series variation comes from changing sentence frequencies c_{jit}.
- This avoids estimating separate language models year by year, which would be noisy in early sample years.

Interpretation. The paper asks whether the mix of finance-related language becomes more or less positive over time. It does not reestimate the meaning of language separately in every year.

2.9 Standardized first difference

Because the level of finance sentiment trends upward over time, the macroeconomic analysis mainly uses first differences:

$$\Delta f_{it} = \frac{f_{it} - f_{it-1}}{\sigma(f_{it})}. \tag{3}$$

The denominator is the full-sample standard deviation of finance sentiment. The authors note that within-country variation after absorbing country fixed effects is much smaller than the unconditional cross-country variation.

Interpretation. In the macro tests, the paper focuses on changes in sentiment rather than permanent cross-country differences. This helps reduce the concern that country-level culture or institutions mechanically drive every result.

2.10 Main descriptive facts

- U.S. and U.K. English have the highest finance sentiment.

- Spanish, French, Chinese, and Italian are in the middle.
- German and Russian have negative average finance sentiment.
- Chinese finance sentiment is volatile, with large changes around the early 1970s, international recognition of the People's Republic of China, Nixon's visit, and later market-oriented reforms.
- Russian finance sentiment is very negative and volatile around the Russian Revolution and Soviet period, but rises after the collapse of the USSR.
- Finance sentiment generally becomes more positive over the long sample, with a mild dip around the 2007–2008 global financial crisis.

Interpretation. The level and evolution of finance sentiment line up with major historical regimes and events, which provides internal validation for the text-based measure.

3 Models and empirical specifications (all equations + interpretation)

3.1 Measurement model

The measurement model has three steps:

1. Embed each finance-mentioning sentence s_{ji} into BERT vector space.
2. Define a language-specific finance positivity vector p_i from positive-minus-negative sentence pairs.
3. Compute cosine similarity a_{ji}, then aggregate by language-year into f_{it}.

Interpretation. The model transforms qualitative text into a numerical sentiment index. The important object is not whether a sentence contains a specific positive or negative word, but whether the sentence's context is semantically close to positive or negative finance concepts.

3.2 Capitalism regressions

To validate whether finance sentiment behaves like a measure of attitudes toward finance, the paper relates it to measures of capitalism:

$$f_{it} = \alpha + \beta C_{it} + \gamma \text{GDPgrowth}_{it} + \mu_i + \lambda_t + \varepsilon_{it}, \tag{4}$$

where C_{it} is one of the following capitalism proxies:

- ideology of the largest party in government, coded from left to center to right;
- survey-based preference for private rather than public ownership of business;
- difficulty of registering a business.

Interpretation. If the measure captures attitudes toward finance, it should be more positive in societies where private enterprise is viewed more favorably and business formation is easier. The estimates confirm this pattern, but the low explanatory power without fixed effects suggests finance sentiment is not simply another label for capitalism.

3.3 External validation regressions

The paper validates the measure by estimating regressions of financial participation and inequality outcomes on finance sentiment:

$$Y_{it} = \alpha + \beta f_{it} + \gamma \text{GDPgrowth}_{it} + \mu_i + \lambda_t + \varepsilon_{it}. \tag{5}$$

Dependent variables include:

- household equity holdings as a percent of total household assets;
- household loans/GDP;
- nonfinancial private-sector loans/GDP;
- income inequality measured by the Gini coefficient.

Interpretation. The coefficient β asks whether, within a country and year environment, higher finance sentiment is associated with more use of financial markets or lower inequality. The answer is yes for participation and credit measures, and negative for inequality.

3.4 Other-industry placebo regressions

The paper constructs analogous sentiment measures for coal, paper, tobacco, and the generic word January.

For tobacco, the paper checks whether tobacco sentiment predicts tobacco consumption:

$$TobaccoUse_{it} = \alpha + \beta TobaccoSentiment_{it} + \mu_i + \lambda_t + \varepsilon_{it}. \tag{6}$$

Interpretation. The tobacco exercise is a method validation. If the embedding approach works, then more positive tobacco sentiment should correspond to higher smoking prevalence and cigarette consumption. The paper finds that it does.

3.5 Crisis event-study regression

The paper studies finance sentiment around banking distress episodes with an event-study specification:

$$\Delta f_{it} = c + \sum_{k=-5}^{5} \beta_k \mathbf{1}\{\text{banking distress in } t-k\} + \mu_i + \lambda_t + \varepsilon_{it}. \tag{7}$$

The event is a banking distress episode in which bank equity declines by at least 30 percent.

Interpretation. This regression asks whether sentiment falls before, during, or after financial crises. The key result is that sentiment falls one year before banking distress but remains relatively flat afterward.

3.6 Crisis-prediction regressions

The paper then estimates whether finance sentiment changes have incremental predictive power for future bank equity declines:

$$\mathbb{P}(\text{Crisis}_{i,t+1} = 1) = \Lambda\left(\alpha_i + \beta_0 \Delta f_{it} + \beta_1 \Delta f_{i,t-1} + \gamma_0 \text{CreditGrowth}_{it} + \gamma_1 \text{CreditGrowth}_{i,t-1}\right). \quad (8)$$

The authors also estimate linear probability versions and add year fixed effects in some specifications.

Interpretation. The coefficient on Δf_{it} measures whether a current decline in finance sentiment predicts a higher probability of banking distress next year, holding credit growth constant. The result is negative and significant: lower sentiment predicts crises.

3.7 Alternative crisis definitions

The paper repeats the crisis-prediction analysis with alternative crisis definitions:

- banking panics;
- bank failures;
- bank equity crises;
- the union of Baron-Verner-Xiong crisis indicators;
- Jorda-Schularick-Taylor crisis chronologies.

Interpretation. The prediction result is robust to several definitions from Baron, Verner, and Xiong, but not to the Jorda-Schularick-Taylor narrative crisis measure. This suggests the timing and definition of crises matter.

3.8 Local projections for macroeconomic growth

To study dynamic macroeconomic responses, the paper estimates local projections:

$$\Delta_h y_{i,t+h} = \alpha_i^h + \sum_{k=1}^{3} \beta_k^h \Delta f_{i,t-k} + \sum_{k=0}^{3} \gamma_k^h X_{i,t-k} + \varepsilon_{i,t+h}, \quad h = 0, \ldots, H. \quad (9)$$

Here $\Delta_h y_{i,t+h} = y_{i,t+h} - y_{i,t-1}$ is cumulative growth through horizon h. The vector X_{it} contains lagged macro variables such as GDP growth, credit growth, and finance sentiment, depending on the response variable.

Interpretation. The local projection traces how output and credit evolve after a shock to finance sentiment. The key result is that positive sentiment shocks are followed by higher output and credit growth.

3.9 Credit-channel local projections

In the six-country credit-data subsample, the authors estimate impulse responses for GDP growth, finance sentiment, and credit growth jointly.

Interpretation. If finance sentiment matters because it affects financial participation or regulation, it should also affect credit. The paper finds that a finance sentiment shock is followed by higher credit growth, and adding credit growth partly reduces the GDP response. This suggests that credit is one channel, but not the whole story.

3.10 Prolonged sentiment expansion regressions

To distinguish short-run sentiment shocks from prolonged expansions, the paper estimates regressions of future GDP growth on cumulative sentiment and debt expansions:

$$\text{GDPgrowth}_{i,t+1} = \alpha_i + \beta \Delta f_{i,t-3\to t} + \gamma_{HH} \Delta d^{HH}_{i,t-3\to t} + \gamma_F \Delta d^{F}_{i,t-3\to t} + \varepsilon_{i,t+1}. \tag{10}$$

Here Δd^{HH} is cumulative household debt/GDP growth over three years and Δd^{F} is cumulative firm debt/GDP growth over three years.

Interpretation. This specification asks whether long sentiment booms behave like credit booms. The finding is more cautionary: prolonged finance sentiment expansions are negatively correlated with future GDP growth, even after controlling for household and firm debt expansions.

4 Identification and instruments (what makes the estimates credible)

4.1 There is no formal instrument

The paper does not present a traditional instrumental-variable design. It is primarily a measurement and macro-historical evidence paper.

- The text-based measure is new and carefully validated.
- The macroeconomic regressions are observational.
- The paper is explicit that the local-projection impulse responses are suggestive of causal effects only under strong assumptions.

Interpretation. The paper's credibility comes from measurement validation, timing, fixed effects, robustness, and placebo exercises, not from random assignment.

4.2 Internal validation of the sentiment measure

The authors validate the measure internally in three ways.

- **Sentence inspection.** The most positive American English sentences are about financial management, financial support, and business experience; the most negative sentences are about financial panic, turmoil, and instability.
- **Historical timing.** Large country-level sentiment movements line up with salient events such as the Panic of 1873, Chinese political and economic shifts in the 1970s, World War II, the Russian Revolution, the Soviet famine, the collapse of the USSR, the Great Depression, and German reparations politics.
- **Cross-language ordering.** The persistent country ordering makes economic sense: English-language countries are most positive, while Russian and German sentiment are more negative.

Interpretation. The measure passes a smell test: it assigns positive and negative scores to sentences and historical episodes in ways that are consistent with economic intuition.

4.3 External validation

External validation uses variables that should plausibly move with positive attitudes toward finance.

- More positive finance sentiment is associated with more household equity participation.
- More positive finance sentiment is associated with higher household loans/GDP and higher nonfinancial private-sector loans/GDP.
- More positive finance sentiment is associated with lower income inequality.

Interpretation. These validation tests do not prove causality, but they show that the text measure correlates with real financial behavior in the expected direction.

4.4 Placebo and falsification logic

The paper constructs sentiment measures for banks, mutual funds, stock markets, large corporations, coal, paper, tobacco, and January.

- Specific financial concepts produce broadly similar country ordering to the finance measure.
- Coal, paper, tobacco, and January do not display the same finance-like ordering or trend.
- Tobacco sentiment predicts smoking prevalence and cigarette consumption, showing that the method can also recover meaningful industry-specific attitudes outside finance.

Interpretation. The placebo evidence reduces the concern that the finance sentiment panel is merely capturing a general trend toward positive language or country-specific optimism.

4.5 Fixed effects and standard errors

- Many validation regressions include country and year fixed effects.
- Country fixed effects absorb persistent cross-country differences such as institutions, culture, and language-level sentiment baselines.

- Year fixed effects absorb common global shocks, such as global crises or worldwide changes in publishing language.
- The authors often use Driscoll-Kraay standard errors to account for cross-sectional and serial correlation in a narrow macro panel.

Interpretation. The fixed effects make the estimates rely more on within-country changes, which is more credible than raw cross-country comparisons. But fixed effects cannot eliminate all time-varying omitted variables.

4.6 Timing evidence around crises

The event-study evidence shows that finance sentiment declines before banking distress rather than after it.

- This timing weakens the simple story that crises cause sentiment declines.
- It supports the possibility that deteriorating sentiment toward finance can contribute to financial fragility.
- Publication lags in books would likely strengthen the lead interpretation, because sentiment appearing in books after a lag may reflect earlier changes in public perception.

Interpretation. The crisis timing is one of the paper's strongest arguments for economic relevance. It is still not definitive causal evidence because sentiment and crises may both be driven by unobserved early warning signals.

4.7 Main threats to identification

- **Reverse causality.** Economic downturns, credit conditions, or political instability may affect finance sentiment rather than the other way around.
- **Omitted variables.** Institutional change, wars, education, ideology, media censorship, and globalization may affect both finance sentiment and economic outcomes.
- **Publication selection.** Books are not a random sample of public opinion. They overrepresent authors, publishers, policymakers, and educated elites.
- **Language-country mapping.** A language is not always a country. Spanish, English, and Russian books may be produced outside Spain, the United States, the United Kingdom, or Russia.
- **Time-invariant embeddings.** The method assumes the BERT-implied meaning of sentence embeddings is stable enough over time, while actual language evolves.
- **Credit-data limits.** Credit regressions exclude China and Russia, which are important countries in the sentiment panel.

Interpretation. The paper is best read as providing strong measurement and suggestive macro-historical evidence, not a fully identified causal estimate of the effect of sentiment on growth.

4.8 What the paper can and cannot distinguish

- **Can show:** finance sentiment is persistent, measurable, historically meaningful, correlated with financial participation, and related to crises and growth dynamics.
- **Can suggest:** public attitudes toward finance may influence credit, financial development, and macroeconomic outcomes.
- **Cannot fully prove:** that sentiment alone causes output growth, credit growth, or banking crises.
- **Cannot fully separate:** whether sentiment affects outcomes through household trust, firm financing, bank behavior, regulation, political ideology, or elite narratives.

Interpretation. The most conservative reading is that finance sentiment is a powerful state variable. The more causal reading is that sentiment changes can affect financial intermediation and macroeconomic activity.

5 Tables and figures

5.1 Table 1: Finance mentions across languages

Read:

- Table 1 lists the finance word stems used in each language and the number of unique and total finance-mentioning 5-grams.
- American English, Chinese, Russian, and French have especially large numbers of unique finance sentences.
- German and Italian have fewer finance mentions, which may imply more measurement noise.

Economic message: The corpus is large enough to create a long-run finance sentiment panel, but sentence coverage differs across languages. The measurement exercise is therefore strongest where finance mentions are abundant.

5.2 Figure 1: Conceptual diagram of finance sentiment measurement

Read:

- Positive benchmark finance sentences cluster on one side of the embedding space, negative benchmark sentences on the other.
- A finance sentence is scored by its projection onto the positive-minus-negative dimension.
- Sentences about finance supporting development are positive; sentences about malpractice or panic are negative.

Table 1
Finance mentions across languages

Language	Finance word stem	Unique sentences	Total sentences
American English	financ	220k	79m
British English	financ	48k	15m
Simplified Chinese	金融, 金融, 金_融	196k	305m
French	financ	100k	43m
German	finanz	28k	7m
Italian	finanz	23k	9m
Russian	финан	187k	250m
Spanish	finan	89k	33m

We report the number of mentions of the word finance, translated, and stemmed, in a five-word sequence (5-gram) for each language in the Google Books Ngram Corpus. Our data set covers the period 1870–2009.

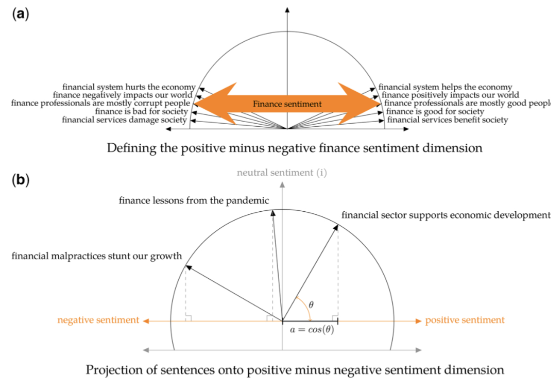

Figure 1
Conceptual diagram of finance sentiment measurement
Panel (a) shows a conceptual diagram of how similar sentences aggregate to a two-dimensional embedding space. We take the vector difference between positive and negative embeddings to define the finance sentiment dimension. Panel (b) illustrates the classification of three example sentences by projecting them onto this dimension. For one of the sentences, we illustrate cosine similarity, defined as the cosine of the angle between two vectors. Sentences that are close in terms of meaning have a smaller angle between them in this vector space, thus higher cosine similarity. Positive finance sentences have a smaller angle to the positive dimension and a larger positive projection on the finance sentiment dimension.

Economic message: The figure explains why the measure is not a dictionary word count. It uses context and semantic distance to quantify whether finance is discussed as useful or harmful.

5.3 Table 3: Positive and negative finance sentences

Read:

- The most positive American English sentences involve financial support, financial management, business experience, and organizational support.
- The most negative sentences involve financial panic, market turmoil, instability, and severe setbacks.

Economic message: The sentence rankings validate the measure at the micro level. The algorithm's most positive and negative examples are consistent with human intuition.

Table 3
Sentences assigned the most positive and negative finance sentiment for American English

Positive sentiment sentences	Negative sentiment sentences
the goal of financial management	turmoil in the financial markets
finance in the graduate school	finances become disordered the
financial support of the center	financial panic swept the country
financial management of the organization	turmoil in financial markets
business and financial experience	financial panic swept the nation
financial support of the graduate	instability in the financial markets
financial support of the science	financial panic in the country
financial support of the course	severe financial setbacks
financial support of the field	a major financial panic
knowledge of the financial structure	world wide financial panic

A sentence is assigned positive or negative finance sentiment based on its projection onto the finance positivity dimension (cosine similarity). Sentences at the top are the most positive or negative in their respective column, and the absolute value of finance sentiment decreases down each list.

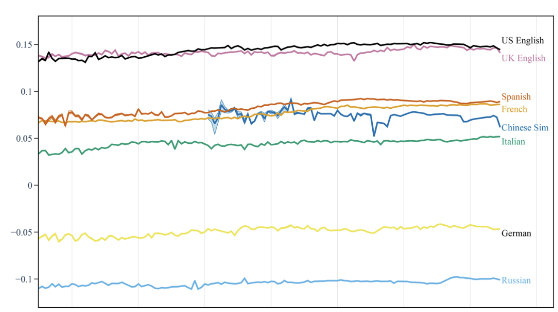

Figure 2
Sentiment toward finance
Finance sentiment is based on the annual average projection of finance-mentioning sentences' embeddings onto the positive minus negative finance sentiment dimension. Sentences are from the Google Books Ngram Corpus and embedded using BERT. Bands represent 95% confidence intervals produced by subsampling.

5.4 Figure 2: Sentiment toward finance

Read:

- U.S. and U.K. English sit at the top of the distribution.
- Spanish, French, Chinese, and Italian are in the middle.
- German and Russian are negative throughout most of the sample.
- The cross-country ordering is persistent, while Chinese sentiment shows large event-related movements.

Economic message: Attitudes toward finance are not merely temporary reactions to recent events. They look like persistent cultural or institutional features, though they can still shift around major shocks.

5.5 Table 4: Finance sentiment and summary statistics

Read:

- The U.S. has the highest mean finance sentiment, followed closely by the U.K.
- Russia and Germany have negative average finance sentiment.
- Total mean GDP growth is about 2.1 percent; total mean credit growth is about 5.9 percent.
- Credit data are unavailable for China and Russia in the authors' macrohistory source.

Economic message: The summary statistics document both the persistent cross-country heterogeneity in finance sentiment and the macro variables used in later tests.

Table 4
Finance sentiment and other summary statistics

Country (language)	Variable, %	Mean	Std. Dev.	Obs.
China	*Finance sentiment*	7.4	0.5	88
(Chinese)	Δ *Finance sentiment*	−0.01	0.57	87
	GDP growth	4.3	7.2	87
France	*Finance sentiment*	7.6	0.7	140
(French)	Δ *Finance sentiment*	0.01	0.09	139
	GDP growth	1.9	6.4	139
	Credit growth	4.9	12.9	101
Germany	*Finance sentiment*	−4.9	0.5	140
(German)	Δ *Finance sentiment*	0.01	0.22	139
	GDP growth	2.1	8.1	139
	Credit growth	8.9	17.8	129
Italy	*Finance sentiment*	4.4	0.4	140
(Italian)	Δ *Finance sentiment*	0.01	0.21	139
	GDP growth	2.0	4.7	139
	Credit growth	6.1	13.9	139
Russia	*Finance sentiment*	−10.4	0.3	140
(Russian)	Δ *Finance sentiment*	0.01	0.18	139
	GDP growth	2.0	8.5	139
Spain	*Finance sentiment*	8.3	0.7	140
(Spanish)	Δ *Finance sentiment*	0.01	0.24	139
	GDP growth	2.1	5.0	139
	Credit growth	7.4	11.1	98
UK	*Finance sentiment*	14.2	0.3	140
(British English)	Δ *Finance sentiment*	0.002	0.21	139
	GDP growth	1.5	2.9	139
	Credit growth	4.0	8.2	129
US	*Finance sentiment*	14.5	0.6	140
(American English)	Δ *Finance sentiment*	0.01	0.18	139
	GDP growth	2.1	5.0	139
	Credit growth	4.5	6.7	129
Total	*Finance sentiment*	5	8.4	1,068
	Δ *Finance sentiment*	0.01	0.25	1,060
	GDP growth	2.1	6.2	1,060
	Credit growth	5.9	12.5	725

The sample spans from 1870 to 2009 for eight country-language pairs. The corpus of sentences for each language is gathered from the Google Books Ngram Corpus. The connotation for each finance-mentioning sentence is measured based on its cosine similarity with respect to the positive minus negative vector. *Finance sentiment* is the average cosine similarity of finance-mentioning sentences in each language and year. GDP and credit data are from Jordà, Schularick, and Taylor (2017) and Barro and Ursua (2010) when available.

Table 5
Finance sentiment and attitudes toward capitalism

	Finance sentiment								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Right-leaning government	0.310***	0.324***	0.00338						
	(0.100)	(0.116)	(0.00447)						
Preference for private ownership of business				0.489***	0.515***	0.00533			
				(0.0327)	(0.0333)	(0.0252)			
Difficulty of registering a business							−0.179***	−0.187***	−0.00261
							(0.00636)	(0.0109)	(0.00218)
GDP growth	0.0231	0.0246	−0.000342	0.116***	0.126***	0.00394*	0.00898	−0.0360	−0.00368
	(0.0243)	(0.0311)	(0.00109)	(0.0147)	(0.0194)	(0.00187)	(0.0302)	(0.248)	(0.00252)
Country FE			Yes			Yes			Yes
Year FE		Yes	Yes		Yes		Yes		Yes
Observations	263	263	263	67	67	67	35	35	35
R^2	0.0293	0.0518	0.999	0.522	0.551	1.000	0.222	0.231	1.000

Right-leaning government is based on the ideology of the largest party in government, according to the classification scheme of the Database of Political Institutions, made available via the World Bank starting in 1975. Scale - Left: 0, Center: 0.5, Right:1. Preference for business ownership is measured using the World Values Surveys, available after 1992. If survey data for a specific year is missing, we estimate the values by linearly extrapolating between the nearest available years. Scale - State: 1, Private: 10. Difficulty of registering a business measures the total number of procedures required for a startup to obtain legal status, obtained from World Development Indicators published by the World Bank starting in 2003. The scale ranges from 5 to 10. Finance sentiment is standardized to have a mean of zero and a standard deviation of one. Driscoll-Kraay standard errors are in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table 6
Credit, stock market participation, and income inequality

	(1) Equity/Total for households (%)	(2) Households loans / GDP	(3) Non-financial private sector loans / GDP	(4) Income inequality
Finance sentiment	57.77*	90.70***	102.6***	−45.42***
	(27.65)	(31.00)	(29.66)	(11.31)
GDP growth	−0.377	0.00873	−0.311	−0.0175
	(0.264)	(0.255)	(0.210)	(0.0944)
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	75	431	733	216
R^2	0.952	0.811	0.718	0.880

Equity as a percent of total assets for households is from the OECD and covers five countries: France, Germany, Spain, the United Kingdom, and the United States. This metric ranges from 0 to 100 and is available from 1995 onward. Total loans to households and total loans to the non-financial private sector, expressed as a percentage of gross domestic product, are sourced from the Jordà-Schularick-Taylor Macrohistory Database. The available years for these metrics vary by country, with data for total loans to households typically starting from 1950 and for total loans to the non-financial private sector from 1900. Income inequality is measured using the Gini coefficient, reported by the World Income Inequality Database, with values ranging from 0 to 100. While the availability of data varies by country, it is generally accessible starting from 1970. Finance sentiment is standardized to have a mean of zero and a standard deviation of one. Driscoll-Kraay standard errors are in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

5.6 Table 5: Finance sentiment and attitudes toward capitalism

Read:

- Right-leaning governments are associated with more positive finance sentiment.
- Survey preference for private ownership of business is associated with more positive finance sentiment.
- Difficulty of registering a business is associated with less positive finance sentiment.
- With country fixed effects, R-squared is near one because the level of finance sentiment is highly persistent across countries.

Economic message: Finance sentiment is related to capitalism and private enterprise, but it is not fully explained by capitalism proxies. It captures something more specific about the social meaning of finance.

5.7 Table 6: Credit, stock market participation, and income inequality

Read:

- Finance sentiment is positively associated with household equity shares.
- It is positively associated with household loans/GDP and nonfinancial private-sector loans/GDP.
- It is negatively associated with income inequality.
- The regressions include country and year fixed effects.

Economic message: The validation results suggest that finance sentiment is connected to real financial participation and to distributional outcomes, not just to language.

5.8 Figures 3 and 4: Specific finance concepts and general sentiment

Read:

- Figure 3 repeats the sentiment method for banks, mutual funds, stock markets, and large corporations.
- The ordering across countries is broadly similar to the main finance sentiment measure.

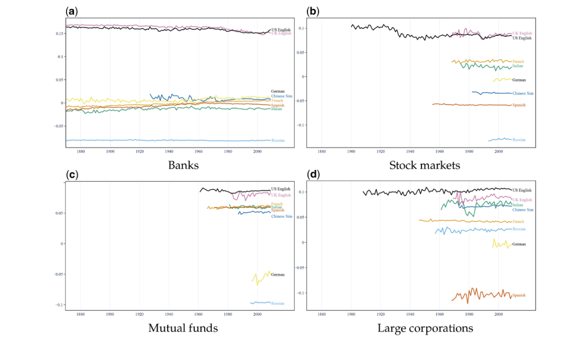

Figure 3
Sentiments toward specific financial concepts
We report sentiment toward banks, mutual funds, stock markets, and large corporations by redefining the positivity dimension accordingly and focusing on sentences mentioning these specific concepts. For example, sentiment for banks is based on the annual average projection of bank-mentioning sentence embeddings onto the positive minus negative bank sentiment dimension. To define the positivity dimension, we average the difference in embedding for the following tuples (and their translations to each language): [("banks are good for society," "banks are bad for society"),("bank managers are mostly good people," "bank managers are mostly corrupt people"),("banks positively impact our world," "banks negatively impact our world"),("banking industry helps the economy," "banking industry hurts the economy"),("banking services benefits society," "banking services damages society")]. For banks, we only report sentiment for languages where the number of sentences containing the word, in a year, exceeds 1,000. For the rest, because the number of sentences in a year is smaller, we use a cutoff equal to the

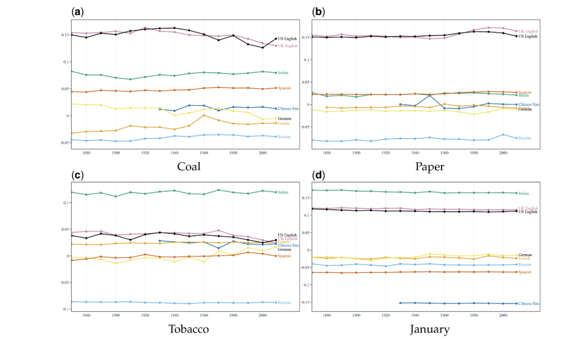

Figure 4
General sentiment
We report sentiment toward the coal, paper and tobacco industries, as well as toward the fairly generic word "January," redefining the positivity dimension accordingly and focusing on sentences mentioning these alternatives instead of "finance." For example, sentiment for the coal industry is based on the annual average projection of coal-mentioning sentence embeddings onto the positive minus negative coal sentiment dimension. To define the positivity dimension, we average the difference in embedding for the following tuples (and their translations to each language): [("coal is good for society," "coal is bad for society"),("coal miners are mostly good people," "coal miners are mostly corrupt people"),("coal positively impacts our world," "coal negatively impacts our world"),("coal industry helps the economy," "coal industry hurts the economy"),("coal industry benefits society," "coal industry damages society")]. Between similar ngrams such as "coal miners," "coal producers," we select the ngram with a higher Google search volume. We only report sentiments for the language when the number of sentences containing

- Figure 4 repeats the method for coal, paper, tobacco, and January.
- General-industry sentiment does not show the same trend or country ordering as finance sentiment.

Economic message: The finance sentiment panel is not just general optimism or language positivity. It captures a finance-specific dimension of social attitudes.

5.9 Table 7: Tobacco sentiment and tobacco consumption

Read:

- Tobacco sentiment is positively associated with daily smoking prevalence.
- Tobacco sentiment is positively associated with annual cigarette consumption per capita.
- A within-country standard deviation increase in tobacco sentiment corresponds to about 0.5 percentage points more daily smoking prevalence and about 83 more cigarettes per capita annually.

Economic message: The text-embedding method can recover economically meaningful attitudes outside finance. This strengthens confidence in the measurement approach.

5.10 Figure 5: Finance sentiment around financial crises

Read:

- Finance sentiment falls one year before banking distress.
- Sentiment is relatively flat after the distress event.
- The pre-crisis decline remains visible when controlling for credit growth and GDP growth, although the sample excludes China and Russia because of missing credit data.

Economic message: The crisis evidence challenges a simple "crises make people dislike finance" story. Instead, worsening finance sentiment appears before banking distress.

Table 7
Tobacco consumption and tobacco sentiment

	(1) Daily smoking prevalence (%)	(2) Consumption per capita (# cig.)
Tobacco Sentiment	5.143***	873.5***
	(1.619)	(132.5)
Country FE	Yes	Yes
Year FE	Yes	Yes
Observations	240	240
R^2	0.902	0.682

Daily smoking prevalence and annual cigarette consumption are from Global Health Data Exchange made available by the Institute for Health Metrics and Evaluation, covering the 1980–2009 period. Annual cigarette consumption is normalized by population data from the World Bank. Tobacco sentiment is standardized to have a mean of zero and a standard deviation of one. Driscoll-Kraay standard errors are in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

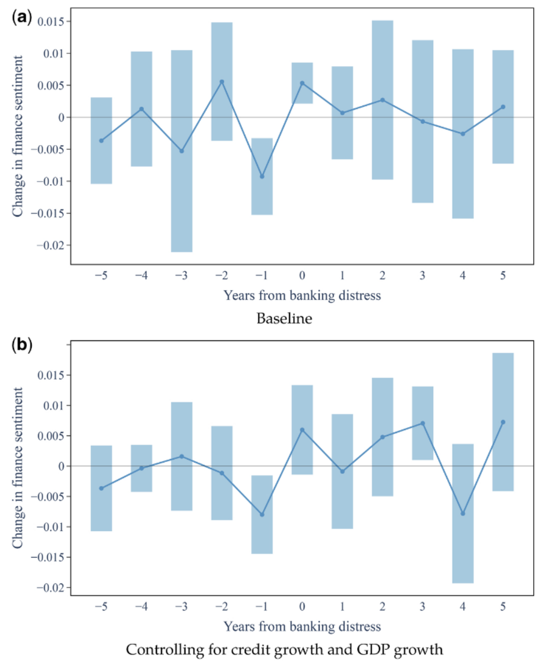

Figure 5
Finance sentiment around financial crises
The figure shows how finance sentiment evolves around periods of banking distress, when bank equity declines by at least 30%. The light blue bars represent 90% confidence intervals adjusted for country-level clustering. Estimates are based on the following regression: $\Delta f_{it} = c + \beta_1 crisis_{it}^{-5} + \cdots + \beta_{10} crisis_{it}^{+5} + country_i + yr_t + \epsilon_{i,t}$. Model specifications with credit growth exclude China and Russia as credit data for these two countries are unavailable. Finance sentiment is standardized to have a mean of zero and a standard deviation of one.

5.11 Table 8: Finance sentiment declines before bank equity declines

Read:

- The coefficient on current Δf_t is negative and statistically significant across the main specifications.
- The result survives controls for credit growth.
- Lagged credit growth is positive, consistent with credit booms preceding banking distress.

Economic message: Finance sentiment has incremental predictive power for crises beyond credit growth. This is one of the paper's most important empirical results.

Table 8
Finance sentiment declines before bank equity declines

	Bank equity 30% decline$_{t+1}$				
Model	Logistic regression		Linear probability model		
	(1)	(2)	(3)	(4)	(5)
Δf_t	−8.518***	−6.853*	−15.62***	−0.967**	−1.135***
	(3.031)	(3.700)	(3.019)	(0.382)	(0.432)
Δf_{t-1}		5.788	5.419	0.168	−0.0549
		(4.394)	(8.601)	(0.329)	(0.325)
Credit growth$_t$			−0.0114	−0.000767	−0.00113
			(0.00867)	(0.000613)	(0.000728)
Credit growth$_{t-1}$			0.0312***	0.00258**	0.00202*
			(0.00925)	(0.00115)	(0.00119)
Country FE	Yes	Yes	Yes	Yes	Yes
Year FE					Yes
Observations	1,052	1,044	709	709	709
R^2	0.00971	0.0129	0.0494	0.03	0.354

We report regressions of banking crisis indicators on lagged finance sentiment changes and credit growth. Banking crises defined in Baron, Verner, and Xiong (2021) represent banking equity declines a 30% for each country in a year. Credit growth is loan growth as defined in Schularick and Taylor (2012). Model specifications with credit growth exclude China and Russia as credit data of these two countries are unavailable. Finance sentiment is standardized to have a mean of zero and a standard deviation of one. Bootstrap robust standard errors are in parentheses for logistic regression models and Driscoll-Kraay standard errors for linear probability models. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table 9
Alternative definitions of banking crises

Logistic regression	Panic$_{t+1}$ (1)	Bank failures$_{t+1}$ (2)	Bank equity crisis$_{t+1}$ (3)	BVX crisis$_{t+1}$ (4)	JST crisis$_{t+1}$ (5)
Δf_t	−15.05**	−15.80*	−18.42*	−14.33**	0.770
	(6.047)	(9.113)	(10.79)	(6.204)	(6.640)
Δf_{t-1}	5.464	7.724	15.82	6.307	15.97
	(22.23)	(24.19)	(16.46)	(20.67)	(9.923)
Credit growth $_t$	−0.0156	−0.0264	−0.0236	−0.0163	−0.0132
	(0.0190)	(0.0175)	(0.0183)	(0.0174)	(0.0151)
Credit growth $_{t-1}$	0.0421**	0.0524**	0.0523***	0.0429**	0.0424*
	(0.0212)	(0.0213)	(0.0136)	(0.0203)	(0.0249)
Country FE	Yes	Yes	Yes	Yes	Yes
Observations	709	709	709	709	709
R^2	0.0730	0.106	0.131	0.0729	0.0662

We report regressions of banking crisis indicators on lagged finance sentiment changes and credit growth. For robustness, we consider several definitions of banking crises defined in Baron, Verner, and Xiong (2021). Credit growth is loan growth as defined in Schularick and Taylor (2012). Model specifications with credit growth exclude China and Russia as credit data of these two countries are unavailable. Finance sentiment is standardized to have a mean of zero and a standard deviation of one. Bootstrap robust standard errors are in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

5.12 Table 9: Alternative definitions of banking crises

Read:

- Finance sentiment declines predict banking panics, bank failures, bank equity crises, and the combined Baron-Verner-Xiong crisis definition.
- The result does not hold for the Jorda-Schularick-Taylor narrative crisis chronology.

Economic message: The crisis result is robust to several crisis definitions but not universal. Dating and defining financial crises is an important source of uncertainty.

5.13 Figure 6: Impulse responses of output and finance sentiment

Read:

- A GDP growth shock mean-reverts over time.
- A finance sentiment shock is followed by higher cumulative GDP growth.
- The paper reports that a 1 percentage point increase in finance sentiment raises GDP growth by roughly 0.3 percentage points around four years out.

Economic message: Short-run improvements in finance sentiment are followed by stronger macroeconomic performance, consistent with the idea that positive attitudes toward finance can support productive intermediation.

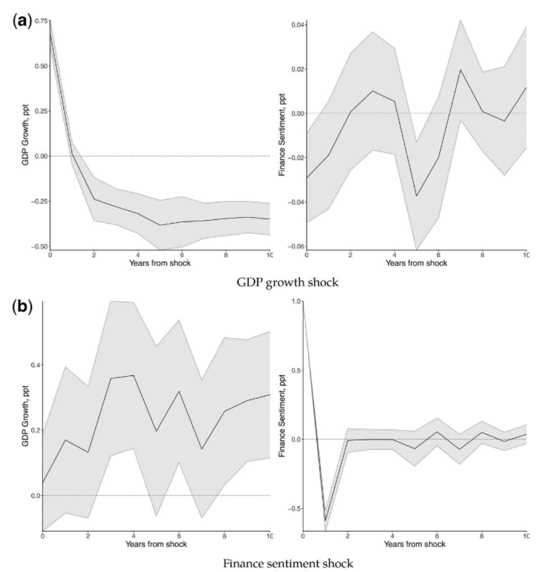

Figure 6
Impulse responses of output and finance sentiment
Impulse responses estimated via local projections indicate the change of the cumulative response to a one percentage point shock. Bands are 90% confidence intervals based on Driscoll and Kraay nonparametric robust standard errors.

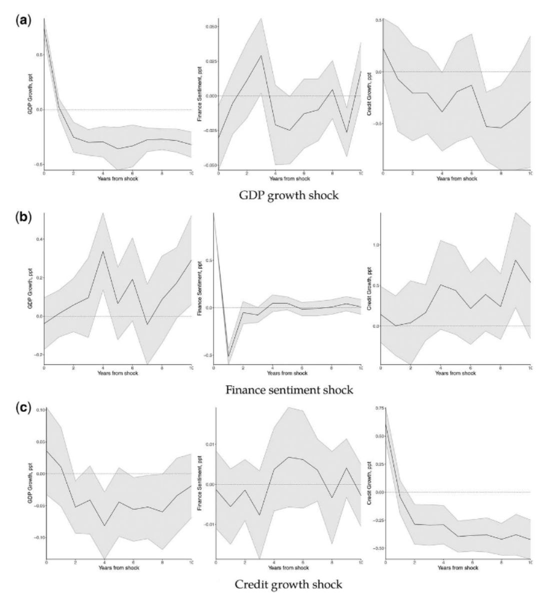

Figure 7
Impulse responses of output, sentiment, and credit (without China and Russia)
Impulse responses estimated via local projections indicate the change of the cumulative response to a 1 percentage point shock. Bands are 90% confidence intervals based on Driscoll and Kraay nonparametric robust standard errors.

5.14 Figure 7: Impulse responses of output, sentiment, and credit

Read:

- The credit-subsample analysis excludes China and Russia.

Table 10
Prolonged sentiment and credit expansions and exchange rate regimes

	GDP growth$_{t+1}$		
Regime	Full sample (1)	Fixed (2)	Flexible (3)
$\Delta f_{\text{t-3}\to\text{t}}$	-15.20^{**}	-18.84^{*}	-18.11^{*}
	(6.44)	(9.77)	(9.48)
$\Delta d^{\text{HH}}_{\text{t-3}\to\text{t}}$	0.03^{***}	0.05^{**}	0.02
	(0.01)	(0.02)	(0.01)
$\Delta d^{\text{F}}_{\text{t-3}\to\text{t}}$	-0.02	-0.03	-0.02
	(0.01)	(0.03)	(0.01)
Country FE	Yes	Yes	Yes
Obs	407	163	188

This table reports regressions of future GDP growth on prolonged expansions of financial sentiment and credit. We consider both household and firm debt as proxy variables for credit expansions. Following Mian, Sufi, and Verner (2017), we define $\Delta d^{\text{HH}}_{\text{t-3}\to\text{t}}$ as the cumulative growth of household debt over GDP over 3 years and $\Delta d^{\text{F}}_{\text{t-3}\to\text{t}}$ as the cumulative growth of firm debt over GDP over 3 years, respectively. Column 1 reports the results for the full sample, while columns 2 and 3 report the results for the fixed exchange regime and the flexible regime (intermediate and freely floating), respectively. All regressions include country fixed effects. Driscoll and Kraay nonparametric robust standard errors, clustered by country, are reported in parentheses.

- A finance sentiment shock is followed by higher GDP growth and higher credit growth.
- The paper reports that a 1 percentage point finance sentiment shock is associated with about a 0.4 percentage point increase in credit growth.

Economic message: Credit appears to be an important channel through which finance sentiment is connected to macroeconomic outcomes, but it does not fully explain the GDP response.

5.15 Table 10: Prolonged sentiment and credit expansions

Table 10
Prolonged sentiment and credit expansions and exchange rate regimes

	GDP growth$_{t+1}$		
Regime	Full sample (1)	Fixed (2)	Flexible (3)
$\Delta f_{\text{t-3}\to\text{t}}$	-15.20^{**}	-18.84^{*}	-18.11^{*}
	(6.44)	(9.77)	(9.48)
$\Delta d^{\text{HH}}_{\text{t-3}\to\text{t}}$	0.03^{***}	0.05^{**}	0.02
	(0.01)	(0.02)	(0.01)
$\Delta d^{\text{F}}_{\text{t-3}\to\text{t}}$	-0.02	-0.03	-0.02
	(0.01)	(0.03)	(0.01)
Country FE	Yes	Yes	Yes
Obs	407	163	188

Future GDP growth after prolonged sentiment and debt expansions.

Read:

- Prolonged finance sentiment expansions are negatively associated with future GDP growth.
- The result is present in the full sample and in fixed and flexible exchange-rate regimes.
- Household debt expansion is positive in some specifications, while firm debt expansion is not significant.

Economic message: A short positive sentiment shock and a prolonged sentiment boom are not the same thing. Sustained optimism toward finance may resemble broader credit-cycle overheating.

6 Short summary

- **Question.** Does society's sentiment toward finance vary over time and across countries, and is that sentiment economically meaningful?
- **Data.** The authors use finance-mentioning five-word sequences from the Google Books Ngram Corpus for eight languages from 1870 to 2009, matched to historical macro-financial data.
- **Measurement.** BERT embeds sentences; cosine similarity to a positive-minus-negative finance dimension gives sentence sentiment; annual finance sentiment is a frequency-weighted average by language-year.
- **Descriptive result.** Finance sentiment is persistently high in U.S. and U.K. English, middle in Spanish/French/Chinese/Italian, and low in German and Russian.
- **Validation.** The measure lines up with capitalism proxies, financial participation, lower inequality, sentence-level inspection, historical events, and placebo industry tests.
- **Crisis result.** Finance sentiment falls one year before bank equity decline episodes and predicts crisis indicators even after controlling for credit growth.
- **Growth result.** Positive finance sentiment shocks are followed by higher output and credit growth in local projections.
- **Caution.** The evidence is not based on a formal causal instrument. Reverse causality and time-varying omitted variables remain possible.
- **Main implication.** Public narratives about finance may matter for financial development, crisis vulnerability, and macroeconomic performance.

7 Conclusion

7.1 Core conclusion

Jha, Liu, and Manela create a new long-run measure of popular sentiment toward finance by applying BERT embeddings to Google Books text. The measure reveals large and persistent cross-country differences: English-language finance sentiment is high, Russian and German sentiment are low, and the middle countries move around major historical events. The measure is externally validated through its connection to financial participation, credit, and inequality.

The paper's central empirical message is that finance sentiment is not merely a cultural curiosity. Declines in finance sentiment predict banking distress, while positive sentiment shocks are followed by higher GDP and credit growth.

7.2 Economic intuition

Finance depends on trust. If households, firms, voters, and policymakers believe finance is useful, they may use financial services more, support financial development, tolerate financial innovation, and allow credit to flow toward productive uses. If they see finance as harmful, corrupt, or predatory, participation and political support may decline, restricting credit, risk sharing, and investment.

This is why public sentiment toward finance can be economically relevant even if it is measured in books. Books reflect the language and narratives through which societies understand finance.

7.3 Takeaway

The paper answers the title question indirectly. Finance may benefit society not only because financial institutions allocate capital, but also because society's beliefs about finance shape whether those institutions are trusted, used, and regulated. The strongest safe conclusion is that finance sentiment is a persistent and economically meaningful macro-financial state variable. The more ambitious interpretation is that improving public sentiment toward finance can itself support financial development and growth.